Chapter 4 Checklist

Trigonometry

Ron Larson and Robert P. Hostetler, *Precalculus with Limits*, 2nd ed., p. 279. Tick each box as you go.

Work through the four parts in order: read each section, practise the matching Duolingo units, work the book-only topics that Duolingo does not cover, then finish the chapter. Duolingo units are grouped by lane (a grade or a topic in the app) and listed in the order the app shows them, so you can follow one lane from top to bottom. **full** = the unit practises the section's main skill, **partial** = part of it, **prereq** = a warm-up from an earlier grade.

In this chapter: If time is short, follow grade 10 for the right-triangle ratios and special-angle values of 4.3, then grade 11 into grade 12 for the sine, cosine, and tangent graphs of 4.5 and 4.6, where Duolingo is strongest. No unit covers radians, the unit circle, reference angles, cosecant, secant, cotangent, or inverse trig functions, so most of 4.1, 4.2, 4.4, and 4.7 comes from the book, with Duolingo giving only single steps and warm-ups. Section 4.8 gets the tools but not the applications: elevation and depression problems, bearings, and harmonic motion models are book only.

Draft. The Duolingo mapping for this chapter has not been reviewed yet, so treat the unit choices as a first pass.

1. Read the textbook sections

- ☐ 4.1 Radian and Degree Measure — p. 280
- ☐ 4.2 Trigonometric Functions: The Unit Circle — p. 292
- ☐ 4.3 Right Triangle Trigonometry — p. 299
- ☐ 4.4 Trigonometric Functions of Any Angle — p. 310
- ☐ 4.5 Graphs of Sine and Cosine Functions — p. 319
- ☐ 4.6 Graphs of Other Trigonometric Functions — p. 330
- ☐ 4.7 Inverse Trigonometric Functions — p. 341
- ☐ 4.8 Applications and Models — p. 351

2. Practise on Duolingo

Pick one lane and work down it; the grade lanes go deepest. The note after each unit gives its rating and the textbook sections it serves.

Grade 4

- ☐ Time *prereq, 4.1* — unit 46
- ☐ Angle measures *prereq, 4.1* — unit 51

Grade 5

- ☐ Add fractions with unlike denominators *prereq, 4.1* — unit 13
- ☐ Subtract fractions with unlike denominators *prereq, 4.1* — unit 14

- ☐ Multiply fractions *prereq, 4.1* unit 21

Grade 6

- ☐ Unit rate *prereq, 4.1* unit 10
- ☐ Unit conversions in the customary system *prereq, 4.1* unit 14
- ☐ Greatest common factor *prereq, 4.1* unit 30
- ☐ Plot points in all quadrants *prereq, 4.1, 4.4* unit 33

Grade 7

- ☐ Circumference of circles *prereq, 4.1, 4.2* unit 41
- ☐ Area of circles *prereq, 4.1* unit 42
- ☐ Complementary angles *partial, 4.1, 4.3, 4.8* unit 43
- ☐ Supplementary angles *partial, 4.1, 4.4* unit 44

Grade 8

- ☐ Alternate interior angles *prereq, 4.8* unit 73
- ☐ Interior angles of a triangle *prereq, 4.8* unit 76

Grade 9

- ☐ Simplify radicals *prereq, 4.3, 4.4* unit 43

Grade 10

- ☐ Define rotations *prereq, 4.1* unit 6
- ☐ 90-degree clockwise rotations *prereq, 4.1* unit 8
- ☐ Proportional relationships in similar triangles *prereq, 4.3* unit 17
- ☐ Pythagorean theorem *prereq, 4.3, 4.4, 4.7, 4.8* unit 20
- ☐ Intro to the sine ratio *full, 4.3, 4.7, 4.8* unit 23
- ☐ Sine of 30 *partial, 4.2, 4.3, 4.4* unit 24
- ☐ Sine of 60 *partial, 4.2, 4.3, 4.4* unit 25
- ☐ Sine of 45 *partial, 4.2, 4.3, 4.4* unit 26
- ☐ Intro to the cosine ratio *full, 4.3, 4.7, 4.8* unit 27
- ☐ Cosine of 60 *partial, 4.2, 4.3, 4.4* unit 28
- ☐ Cosine of 30 *partial, 4.2, 4.3, 4.4* unit 29
- ☐ Cosine of 45 *partial, 4.2, 4.3, 4.4* unit 30
- ☐ Intro to the tangent ratio *full, 4.3, 4.7, 4.8* unit 31
- ☐ Tangent of 45 *partial, 4.2, 4.3, 4.4* unit 32
- ☐ Tangent of 30 *partial, 4.2, 4.3, 4.4* unit 33

- ☐ Tangent of 60 *partial, 4.2, 4.3, 4.4* unit 34
- ☐ Connect sine and cosine *partial, 4.3, 4.7* unit 35
- ☐ Connect sine, cosine, and tangent *partial, 4.2, 4.3, 4.6, 4.7* unit 36
- ☐ Reflections across the x-axis as functions *prereq, 4.2* unit 41
- ☐ Reflections across y=x as functions *prereq, 4.7* unit 43
- ☐ Circles on the plane *prereq, 4.2* unit 47
- ☐ Distance formula *prereq, 4.4* unit 51

Grade 11

- ☐ Exponential decay *prereq, 4.6* unit 42
- ☐ Horizontal translations *prereq, 4.5* unit 70
- ☐ X-axis function reflections *partial, 4.5, 4.6* unit 71
- ☐ Y-axis function reflections *partial, 4.5* unit 72
- ☐ Intro to the sine function *partial, 4.2, 4.5, 4.6, 4.7* unit 77
- ☐ Period of sine functions *partial, 4.2, 4.5* unit 78
- ☐ Intro to the cosine function *partial, 4.2, 4.5, 4.6, 4.7* unit 79
- ☐ Period of cosine functions *partial, 4.2, 4.5* unit 80
- ☐ Midline of sine and cosine functions *full, 4.5* unit 81
- ☐ Amplitude of sine and cosine functions *full, 4.5* unit 82
- ☐ Intro to the tangent function *partial, 4.2, 4.6, 4.7* unit 83
- ☐ Period of tangent functions *full, 4.6* unit 84

Grade 12

- ☐ Vertical asymptotes of rational functions *prereq, 4.6* unit 9
- ☐ Function composition *partial, 4.7* unit 13
- ☐ Inverse functions *partial, 4.7* unit 14
- ☐ Domain and range *partial, 4.7* unit 17
- ☐ Graphing inverses *partial, 4.7* unit 18
- ☐ Period of sinusoidal functions *full, 4.5, 4.6, 4.8* unit 19
- ☐ Midline of sinusoidal functions *full, 4.5* unit 20
- ☐ Amplitude of sinusoidal functions *full, 4.5, 4.6, 4.8* unit 21
- ☐ Phase shift of sinusoidal functions *full, 4.5, 4.6* unit 22
- ☐ Relate sine and cosine *partial, 4.5, 4.8* unit 23

Geometry

- ☐ Circles *prereq, 4.1* unit 12
- ☐ Angle relationships *partial, 4.1* unit 13
- ☐ The Pythagorean theorem *prereq, 4.3, 4.8* unit 23
- ☐ Sine ratio *full, 4.3, 4.8* unit 24
- ☐ Cosine ratio *full, 4.3, 4.8* unit 25
- ☐ Tangent ratio *full, 4.3, 4.8* unit 26
- ☐ Sine function *partial, 4.2, 4.5* unit 27
- ☐ Cosine function *partial, 4.2, 4.5* unit 28
- ☐ Tangent function *partial, 4.2, 4.6* unit 29
- ☐ Graph sine and cosine *full, 4.5, 4.6, 4.8* unit 30

Algebraic Graphing

- ☐ Inverse linear functions *prereq, 4.7* unit 23

3. Section by section

For each section: what Duolingo covers (the units are ticked off in part 2) and the book-only topics to work in the textbook. Tick each book-only topic once you have worked examples of it.

4.1 Radian and Degree Measure p. 280

Duolingo has no unit on standard position, coterminal angles, radians, arc length, angular speed or sectors, so most of this section is book only. The grade 7 complementary and supplementary units practise that skill in degrees, with the Geometry topic's Angle relationships as the Topics-tab alternative. The angle, rotation, quadrant, fraction, factor, circle, conversion, time and rate units are warm-ups for the section's definitions and formulas.

Duolingo: **Grade 4:** 46. Time (prereq); 51. Angle measures (prereq)
Grade 5: 13. Add fractions with unlike denominators (prereq); 14. Subtract fractions with unlike denominators (prereq); 21. Multiply fractions (prereq)
Grade 6: 10. Unit rate (prereq); 14. Unit conversions in the customary system (prereq); 30. Greatest common factor (prereq); 33. Plot points in all quadrants (prereq)
Grade 7: 41. Circumference of circles (prereq); 42. Area of circles (prereq); 43. Complementary angles (partial); 44. Supplementary angles (partial)
Grade 10: 6. Define rotations (prereq); 8. 90-degree clockwise rotations (prereq)
Geometry: 12. Circles (prereq); 13. Angle relationships (partial)

Book only:

- ☐ Angles in standard position: vertex, initial side, terminal side
- ☐ Positive (counterclockwise) and negative (clockwise) angles
- ☐ Angles beyond one full revolution
- ☐ Quadrant of an angle's terminal side
- ☐ Coterminal angles in degrees and radians
- ☐ Definition of radian measure (arc length over radius)

- ☐ Radian measures of common angles and quadrant boundaries
- ☐ Acute and obtuse angles in radians (0 to pi/2, pi/2 to pi)
- ☐ Converting between degrees and radians
- ☐ Degrees-minutes-seconds and decimal degrees
- ☐ Complementary and supplementary angles in radians
- ☐ Arc length formula s = r theta
- ☐ Linear speed v = s/t, angular speed omega = theta/t, and v = r omega
- ☐ Converting revolutions per minute to radians per minute
- ☐ Area of a circular sector

4.2 Trigonometric Functions: The Unit Circle p. 292

No unit is named for radians, the unit circle, or cosecant, secant, and cotangent, so much of this section is book only. Drill the exact first-quadrant values with the Grade 10 special-angle units, which work in degrees (match 30, 45, 60 to pi/6, pi/4, pi/3 yourself), and use Grade 11 for sine, cosine, and tangent as functions and for reading the period; the Geometry topic's three function units are an alternative for the first part only. Book only, among others listed in the gaps: defining all six functions from (x, y), values outside the first quadrant, even and odd functions, using the period to evaluate, and calculator work.

Duolingo: **Grade 7:** 41. Circumference of circles (prereq)
Grade 10: 24. Sine of 30 (partial); 25. Sine of 60 (partial); 26. Sine of 45 (partial); 28. Cosine of 60 (partial); 29. Cosine of 30 (partial); 30. Cosine of 45 (partial); 32. Tangent of 45 (partial); 33. Tangent of 30 (partial); 34. Tangent of 60 (partial); 36. Connect sine, cosine, and tangent (partial); 41. Reflections across the x-axis as functions (prereq); 47. Circles on the plane (prereq)
Grade 11: 77. Intro to the sine function (partial); 78. Period of sine functions (partial); 79. Intro to the cosine function (partial); 80. Period of cosine functions (partial); 83. Intro to the tangent function (partial)
Geometry: 27. Sine function (partial); 28. Cosine function (partial); 29. Tangent function (partial)

Book only:

- ☐ The unit circle and wrapping the real number line around it
- ☐ Defining the six trigonometric functions from the point (x, y) for a real number t
- ☐ Cosecant, secant, and cotangent values as reciprocals
- ☐ Special-angle values in radians (pi/6, pi/4, pi/3), matched to the degree drills
- ☐ Values at the quadrantal points 0, pi/2, pi, 3pi/2
- ☐ Values and signs at standard points outside the first quadrant
- ☐ Where tangent, cotangent, secant, and cosecant are undefined
- ☐ Domain (all reals) and range [-1, 1] of sine and cosine
- ☐ Using the 2pi period to evaluate at large or negative t
- ☐ Even and odd trigonometric functions (cosine and secant even, the others odd)
- ☐ Calculator evaluation of trig functions in radian mode, including csc, sec, cot as reciprocals
- ☐ Estimating sine and cosine values from a unit circle figure, and approximating t for a given value
- ☐ Evaluating trigonometric models at given times, such as a spring's displacement

4.3 Right Triangle Trigonometry p. 299

Grade 10 covers the core in degrees: sine, cosine and tangent as right-triangle ratios, their exact values at 30, 45 and 60 degrees, sine and cosine of complementary angles, and tangent as sine over cosine; the Geometry topic tab is a shorter alternative for the three ratios. Cosecant, secant and cotangent, most identity work, radian forms of the special angles, calculator work, solving right triangles, and angle of elevation and depression problems are book only.

Duolingo: **Grade 7:** 43. Complementary angles (prereq)
Grade 9: 43. Simplify radicals (prereq)
Grade 10: 17. Proportional relationships in similar triangles (prereq); 20. Pythagorean theorem (prereq); 23. Intro to the sine ratio (full); 24. Sine of 30 (full); 25. Sine of 60 (full); 26. Sine of 45 (full); 27. Intro to the cosine ratio (full); 28. Cosine of 60 (full); 29. Cosine of 30 (full); 30. Cosine of 45 (full); 31. Intro to the tangent ratio (full); 32. Tangent of 45 (full); 33. Tangent of 30 (full); 34. Tangent of 60 (full); 35. Connect sine and cosine (partial); 36. Connect sine, cosine, and tangent (partial)
Geometry: 23. The Pythagorean theorem (prereq); 24. Sine ratio (full); 25. Cosine ratio (full); 26. Tangent ratio (full)

Book only:

- ☐ Cosecant, secant, and cotangent as right-triangle ratios
- ☐ Reciprocal identities
- ☐ Quotient identity for cotangent (cot = cos/sin)
- ☐ Pythagorean identities
- ☐ Cofunction identities beyond the sine-cosine pair (tangent-cotangent, secant-cosecant)
- ☐ Special-angle values in radians (pi/6, pi/4, pi/3), and finding the angle from a value
- ☐ Finding all six function values from one given ratio
- ☐ Using fundamental identities to simplify or verify trig expressions
- ☐ Calculator evaluation of trig functions at non-special acute angles (degree mode, decimal degrees and DMS, reciprocals via 1/x)
- ☐ Finding an acute angle from a known side ratio with a calculator
- ☐ Solving right triangles at non-special angles (missing sides)
- ☐ Angles of elevation and depression in applied height and distance problems

4.4 Trigonometric Functions of Any Angle p. 310

Duolingo has no unit on angles in standard position, quadrant signs, or reference angles, so the section's method is book only. It supports single steps: the distance formula, Pythagorean theorem, and radical simplification give r or a missing coordinate, supplementary angles give a second-quadrant reference angle, and the grade 10 special-angle units give the value at a 30, 45, or 60 degree reference angle.

Duolingo: **Grade 6:** 33. Plot points in all quadrants (prereq)
Grade 7: 44. Supplementary angles (prereq)
Grade 9: 43. Simplify radicals (prereq)
Grade 10: 20. Pythagorean theorem (prereq); 24. Sine of 30 (partial); 25. Sine of 60 (partial); 26. Sine of 45 (partial); 28. Cosine of 60 (partial); 29. Cosine of 30 (partial); 30. Cosine of 45 (partial); 32. Tangent of 45 (partial); 33. Tangent of 30 (partial); 34. Tangent of 60 (partial); 51. Distance formula (prereq)

Book only:

- ☐ Six trig functions defined from a point (x, y) and r

- ☐ Cosecant, secant, and cotangent of any angle
- ☐ Signs of the trig functions by quadrant
- ☐ Identifying the quadrant from the signs of two functions
- ☐ Choosing a point on a terminal side given as a line in a stated quadrant
- ☐ Values at quadrantal angles, including where a function is undefined
- ☐ Finding the remaining function values from one value and its quadrant
- ☐ Finding reference angles in degrees and radians
- ☐ Coterminal step for negative angles and angles past one revolution
- ☐ Combining a reference-angle value with the quadrant sign
- ☐ Special-angle values in radians (pi/6, pi/4, pi/3)
- ☐ Finding both angles in one revolution with a given special value
- ☐ Trig functions of real numbers evaluated with reference angles
- ☐ Calculator evaluation of trig functions in degree and radian mode, including csc, sec, cot as reciprocals
- ☐ Evaluating trig models in applied problems

4.5 Graphs of Sine and Cosine Functions p. 319

Grade 11 introduces the sine and cosine graphs with period, midline, and amplitude, and grade 12 repeats them for full sinusoids and adds phase shift, so do grade 11 first, or use the Geometry topic's three sine and cosine units as a shorter alternative. No unit reflects a sinusoid by name, so the general grade 11 reflection units stand in, and the grade 11 horizontal translation unit is a warm-up for phase shift. Key-point sketching, rewriting a negative b with the even/odd properties, frequency, writing an equation from a graph, and fitting sinusoidal models to data are book only.

Duolingo: **Grade 11:** 70. Horizontal translations (prereq); 71. X-axis function reflections (partial); 72. Y-axis function reflections (partial); 77. Intro to the sine function (full); 78. Period of sine functions (full); 79. Intro to the cosine function (full); 80. Period of cosine functions (full); 81. Midline of sine and cosine functions (full); 82. Amplitude of sine and cosine functions (full)
Grade 12: 19. Period of sinusoidal functions (full); 20. Midline of sinusoidal functions (full); 21. Amplitude of sinusoidal functions (full); 22. Phase shift of sinusoidal functions (full); 23. Relate sine and cosine (partial)
Geometry: 27. Sine function (full); 28. Cosine function (full); 30. Graph sine and cosine (full)

Book only:

- ☐ Hand sketching a transformed sinusoid: finding the one-cycle interval and the five key points at quarter-period spacing
- ☐ Rewriting a negative b with the even/odd properties of sine and cosine
- ☐ Frequency as the reciprocal of the period
- ☐ Writing a complete sinusoid equation from its graph (choosing sine or cosine form and combining a, b, c, d)
- ☐ Sinusoidal models fitted to real data (finding a, b, c, d from a table or a description)

4.6 Graphs of Other Trigonometric Functions p. 330

Grade 11 covers the parent tangent graph and its period; prefer that pair over the Topics-tab Geometry unit. The other units are warm-ups for guide curves, transformations, asymptotes and damping factors. Shifts, stretches, and asymptote locations of transformed tangent graphs, cotangent, secant, cosecant, damped graphs and the six-function summary are book only, and grade 12 "Secant lines" is calculus, not the secant function.

Duolingo: **Grade 10:** 36. Connect sine, cosine, and tangent (prereq)
Grade 11: 42. Exponential decay (prereq); 71. X-axis function reflections (partial); 77. Intro to the sine function (prereq); 79. Intro to the cosine function (prereq); 83. Intro to the tangent function (full); 84. Period of tangent functions (full)
Grade 12: 9. Vertical asymptotes of rational functions (prereq); 19. Period of sinusoidal functions (prereq); 21. Amplitude of sinusoidal functions (prereq); 22. Phase shift of sinusoidal functions (prereq)
Geometry: 29. Tangent function (full); 30. Graph sine and cosine (prereq)

Book only:

- ☐ Cotangent graph, period, and asymptotes
- ☐ Secant and cosecant graphs sketched from cosine and sine guide curves
- ☐ Asymptotes and local maxima and minima of secant and cosecant branches
- ☐ Vertical stretch of tangent and cotangent graphs, which have no amplitude
- ☐ Locating two consecutive asymptotes of transformed tangent and cotangent graphs by solving bx - c = -pi/2 and pi/2, or 0 and pi
- ☐ Phase shifts of tangent, cotangent, secant, and cosecant graphs
- ☐ Even/odd symmetry seen in the tangent, cotangent, secant, and cosecant graphs
- ☐ Domain, range, and period summary of all six trigonometric functions
- ☐ Solving equations such as tan x = c or sec x = c graphically on an interval
- ☐ Damped trigonometric graphs and damping factors
- ☐ Bounding curves y = plus or minus f(x) for products such as x sin x
- ☐ Behavior of functions such as (sin x)/x as x approaches 0

4.7 Inverse Trigonometric Functions p. 341

No Duolingo unit teaches arcsin, arccos, or arctan, so the inverse trig functions themselves are book only. Prefer the grade 12 inverse-function block for the general mechanics, with the grade 10 right-triangle ratios and grade 11 trig graphs as warm-ups and the Topics-tab Inverse linear functions unit as a lighter linear-only start. Reading the grade 10 special-angle units from 4.3 (Sine of 30 and so on) backwards gives only acute angles in degrees for positive inputs; radian answers, negative inputs, and choosing the angle inside the restricted range are book only.

Duolingo: **Grade 10:** 20. Pythagorean theorem (prereq); 23. Intro to the sine ratio (prereq); 27. Intro to the cosine ratio (prereq); 31. Intro to the tangent ratio (prereq); 35. Connect sine and cosine (prereq); 36. Connect sine, cosine, and tangent (prereq); 43. Reflections across y=x as functions (prereq)
Grade 11: 77. Intro to the sine function (prereq); 79. Intro to the cosine function (prereq); 83. Intro to the tangent function (prereq)
Grade 12: 13. Function composition (partial); 14. Inverse functions (partial); 17. Domain and range (partial); 18. Graphing inverses (partial)
Algebraic Graphing: 23. Inverse linear functions (prereq)

Book only:

- ☐ Restricting sine, cosine, and tangent to one-to-one intervals
- ☐ Definitions, domains, and ranges of arcsin, arccos, and arctan
- ☐ The notation sin^-1 x and why it is not the reciprocal 1/sin x
- ☐ Exact values of arcsin, arccos, and arctan in radians, including negative and quadrantal inputs
- ☐ Calculator evaluation of inverse trig functions
- ☐ Graphs of y = arcsin x, y = arccos x, and y = arctan x, including transformed versions
- ☐ Inverse properties such as sin(arcsin x) = x and arcsin(sin y) = y, with their interval conditions
- ☐ Compositions whose input lies outside the restricted domain or range
- ☐ Exact values of compositions like tan(arccos a) at a number a, from a right-triangle sketch, including negative a
- ☐ Rewriting compositions such as sin(arctan x) as algebraic expressions in x
- ☐ Writing a triangle angle as an inverse trig function of x
- ☐ Rewriting one inverse trig expression as another via a right triangle (exercise extension)
- ☐ Inverse cotangent, secant, and cosecant (exercise extension)
- ☐ Identities such as arcsin(-x) = -arcsin x and arcsin x + arccos x = pi/2
- ☐ Applied problems that find an angle with an inverse trig function

4.8 Applications and Models

p. 351

Duolingo gives the tools, not the applications: the grade 10 right-triangle ratios (special angles only) support solving right triangles, and the grade 12 period, amplitude, and relate-sine-and-cosine units support reading harmonic motion models. Prefer the grade 10 and grade 12 lanes; the Geometry topic (The Pythagorean theorem through Tangent ratio, then Graph sine and cosine) is the Topics-tab alternative for both halves. Calculator and inverse-trig solving, elevation and depression problems, bearings, and harmonic motion itself (writing and evaluating models, frequency) are book only.

Duolingo: **Grade 7:** 43. Complementary angles (prereq)
Grade 8: 73. Alternate interior angles (prereq); 76. Interior angles of a triangle (prereq)
Grade 10: 20. Pythagorean theorem (prereq); 23. Intro to the sine ratio (partial); 27. Intro to the cosine ratio (partial); 31. Intro to the tangent ratio (partial)
Grade 12: 19. Period of sinusoidal functions (partial); 21. Amplitude of sinusoidal functions (partial); 23. Relate sine and cosine (partial)
Geometry: 23. The Pythagorean theorem (prereq); 24. Sine ratio (partial); 25. Cosine ratio (partial); 26. Tangent ratio (partial); 30. Graph sine and cosine (partial)

Book only:

- ☐ Solving right triangles at non-special angles with a calculator
- ☐ Finding acute angles with inverse trig functions
- ☐ Angles of elevation and depression in applied height and distance problems
- ☐ Two-triangle problems, such as two angles of elevation to the same object
- ☐ Bearings in navigation (N/S-E/W compass form and three-digit nautical form)
- ☐ Simple harmonic motion models d = a sin(wt) and d = a cos(wt)
- ☐ Frequency of harmonic motion (w / 2pi)

- ☐ Writing a harmonic motion model from displacement at t = 0, amplitude, and period (choosing sine or cosine)
- ☐ Evaluating a harmonic motion model at a given time, including the least positive time when d = 0

4. Finish the chapter

- ☐ Chapter Summary p. 362
- ☐ Review Exercises p. 364
- ☐ Chapter Test p. 367
- ☐ Proofs in Mathematics p. 368
- ☐ Problem Solving p. 369

Duolingo unit numbers are each unit's position in that grade's or topic's path in the Duolingo Math app, as recorded on 2026-10-07 (iOS); the app may reorder them. Textbook: Ron Larson and Robert P. Hostetler. Precalculus with Limits, 2nd ed. (Pre-AP National Student Edition) Thomson Brooks/Cole, 2009. ISBN 978-0-538-73659-6. Only titles and unit names are reproduced, for reference; no textbook or Duolingo content is included. This checklist is licensed CC BY 4.0; see the repository NOTICE for third-party material.